Aravindkumar K S

AI / ML ENGINEER

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PROFESSIONAL SUMMARY

AI/ML engineer who has built production RAG and LLM systems end to end in industry - from document ingestion, retrieval and evaluation through inference APIs to the AWS infrastructure that runs them - and went on to formalize that work with an MSc in Mathematical Modelling and Machine Learning (First Class Honours). Proficient in Python, FastAPI and LangChain, with hands-on AWS and Terraform experience. Takes an evaluation-driven approach, using RAGAS and Langfuse to catch regressions and ship reliable, low-risk releases, and applies solid security and compliance practices (least-privilege IAM, PII redaction, GDPR). An AWS Certified Machine Learning Engineer, grounded in a mathematical and research foundation.

WORK EXPERIENCE

Independent Projects & Professional Development

Oct 2025 - Present

AI/ML Engineering

Cork, Ireland

- Designed and built production-grade AI systems to deepen applied LLM and MLOps expertise - an agentic RAG assistant, a self-hosted vLLM inference service, and a CI-integrated LLM evaluation harness (detailed under Projects).

LTIMindtree (now LTM)

Oct 2022 - Aug 2024

Senior Consultant - Package Implementation, Insurance R&D

Bengaluru, India

- Led the proof-of-concept and co-built a production insurance RAG assistant (LangChain, FastAPI), cutting bulk policy-document processing time **~30%** including manual extraction-quality review, at a 4-5s p95 time-to-first-token.
- Owned an event-driven extraction pipeline (Textract, S3, SQS with dead-letter queues, Lambda consumers, Step Functions) parsing **20+ vendor formats** into a Neo4j knowledge-graph insurance ontology at **95%+ accuracy**, auto-quarantining malformed inputs.
- Co-built Studio, the team's central model portal (React, FastAPI) providing model access, management and governance - rate limiting, PII redaction, content moderation, domain gating and application-based access control.
- Independently built a VS Code extension on Studio bringing LLM chat, code generation, review and editing, model switching and history into the IDE for the engineering team.
- Contributed to the LLM inference gateway behind Studio (FastAPI edge, internal gRPC/protobuf), adding a Redis semantic cache that cut redundant LLM calls **~5%** and helping enforce GDPR-compliant PII handling.
- Contributed to evaluation baselines using a RAGAS harness and Langfuse tracing, wiring context-relevance and faithfulness checks into CI as release gates to catch retrieval regressions.
- Contributed to migrating infrastructure-as-code from CloudFormation to Terraform, building reusable ECS/ECR/Lambda/S3 modules with least-privilege IAM and encryption, and cutting AWS spend **~10%** through ECS right-sizing and autoscaling.
- Contributed to containerized blue-green ECS releases via GitHub Actions and CodeBuild, with SonarQube and Trivy security gates enabling low-risk deployments across Dev, Test and Prod.

LTIMindtree (now LTM)

July 2022 - Oct 2022

Graduate Engineer Trainee

Bengaluru, India

- Built Angular tooling and PostgreSQL data models (Spring Boot, Hibernate) for configuring data-pipeline scan jobs and tracking scan lineage, plus SQL aggregations powering data-discovery and coverage dashboards.

PROJECTS

Agentic RAG & Conversational Knowledge Assistant

Built an agentic RAG assistant where a LangChain tool-calling agent routes queries across vector search, SQL lookup and web search, returning Pydantic-validated structured outputs over multi-turn Redis-backed conversation state. Added an LLM-based guardrail layer to catch

ungrounded responses.

Python, Gradio, FastAPI, LangChain, OpenAI, Pinecone, Redis, Pydantic, RAGAS

Optimized LLM Inference Serving

Deployed a self-hosted LLM inference service with vLLM behind a FastAPI gateway, using continuous batching and PagedAttention to raise throughput **~3x** over a naive Hugging Face pipeline.

Applied INT8/AWQ quantization and KV-cache tuning to cut GPU memory **~50%** and p95 latency **~40%**; exposed Prometheus metrics to drive autoscaling and catch performance regressions.

Python, vLLM, FastAPI, PyTorch, Hugging Face Transformers, Docker, Prometheus

LLM Evaluation Harness

Built a reusable LLM evaluation harness scoring models and prompts across accuracy, faithfulness, relevance and latency, with pluggable metrics and provider-agnostic adapters; calibrated LLM-as-judge prompts against human labels.

Wired the harness into CI as regression gates via GitHub Actions, blocking merges on score drops and logging per-run traces to Langfuse.

Python, pytest, RAGAS, DeepEval, Langfuse, OpenAI/Anthropic APIs, GitHub Actions

Wendland's CSRBF in Kolmogorov-Arnold Networks (KAN) Research - UCC

Replaced B-spline and Gaussian edge activations with compactly supported Wendland radial basis functions, matching FC-KAN accuracy on MNIST (97.6% vs 97.7%) with **16% fewer parameters** and **21-23% faster training**, and improving Fashion-MNIST accuracy from 86.7% to 88.7% across six baselines.

PyTorch

EDUCATION

MSc in Mathematical Modelling and Machine Learning University College Cork — Sept 2024 - Sept 2025

Cork, Ireland

First Class Honours (1.1). Coursework: Deep Learning (ANN, RNN, VAE, Transformers, RL), Statistical Methods.

BE in Mechanical Engineering Kumaraguru College of Technology — July 2018 - June 2022

Coimbatore, India

First Class with Distinction (8.8/10.0 CGPA). Publication: "Design Optimization of Grids in Grid Fins for Spacecrafts", International Journal of Vehicle Structures and Systems (IJVSS), 2022.

CERTIFICATIONS

AWS Certified Machine Learning Engineer - Associate	Amazon Web Services	Sept 2026
Model Context Protocol: Advanced Topics	Anthropic	April 2026
Introduction to Model Context Protocol	Anthropic	March 2026

SKILLS

AI & LLM Engineering: RAG & Agentic RAG, LangChain, tool/function calling, prompt engineering, LLM guardrails (PII redaction, content moderation, rate limiting), structured outputs (Pydantic), LLM inference serving (vLLM, continuous batching, PagedAttention, KV-cache, INT8/AWQ quantization), semantic caching, OpenAI & Anthropic APIs, Model Context Protocol (MCP)

Retrieval & Data: Vector search (Pinecone, Chroma), hybrid search, graph retrieval (Neo4j), embeddings, chunking strategies, Amazon Textract, Redis, PostgreSQL, DynamoDB

Coding Agents & Agentic Development: Claude Code, agent skills & plugins, hooks, subagents, MCP servers & tool integration, Docker-sandboxed agent execution, agentic coding workflows

Cloud, MLOps & Security: AWS (ECS, ECR, Lambda, API Gateway, S3, SQS & DLQ, DynamoDB, Step Functions, CloudWatch, SNS, KMS), Terraform, CloudFormation, Docker, CI/CD (GitHub Actions, CodeBuild), MLflow, RAGAS, Langfuse, DeepEval, CI regression gates, SonarQube, Trivy, least-privilege IAM, GDPR compliance, Prometheus, Grafana

Machine Learning: Deep learning (ANN, RNN, VAE, Transformers, RL), Kolmogorov-Arnold Networks, statistical modeling, EDA, feature engineering, Optuna, SMOTE, SHAP, model evaluation & benchmarking

Languages & Frameworks: Python, FastAPI, React, SQL, PyTorch, scikit-learn, XGBoost, Java, JavaScript/TypeScript, Spring Boot, Angular